Lecture-4

HTML Notes: Images, Paths & Semantic Tags

Agenda

1. Images in HTML
2. Path Theory
3. Semantic HTML

Images in HTML

Image Tag Basics

The `<img>` tag is a **self-closing (void) tag** – it only has an opening tag and does not wrap content.

```
<img src="image_path.jpg">
```

Key Characteristics:

- Takes required space equal to the default height and width of the image
- You can specify custom height and width
- Uses paths to locate the image file

Image Tag Attributes:

Attribute	Purpose	Example
`src`	Source path of the image (required)	`src="./photo.jpg"`
`alt`	Alternative text for accessibility	`alt="A beautiful sunset"`
`height`	Height in pixels	`height="200px"`
`width`	Width in pixels	`width="200px"`

Code Example:

```
<h2>Image Tag</h2>

<!-- Local Image -->
```

```
<h3>Local Image</h3>
<img src="./images.jpeg" height="200px" width="200px">

<!-- URL Image -->
<h3>URL Image</h3>
<img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQAPf_062JKWOBQr9rKxfy

<!-- Image Not Loaded (shows alt text) -->
<h4>Image Not Loaded</h4>
<img src="../class_2/images3.jpeg" alt="feather">
```

Clickable Images

You can wrap an image inside an anchor tag to make it clickable.

```
<a href="https://www.youtube.com/" target="_blank">
    <img src="./google.png" height="500px">
</a>
```

How it works: When a user clicks the image, they navigate to the URL specified in the `href` attribute.

Image Types and Use Cases

Format	Use Case	Characteristics
JPEG (.jpg, .jpeg)	Normal images with backgrounds	Pixelates on larger screens, good for photographs
PNG (.png)	Images requiring transparency	Supports transparent backgrounds, pixelates on larger screens
SVG (.svg)	Logos, icons, illustrations	Scales automatically to any screen size, heavier computationally

Code Example:

```
<!-- JPEG - Standard images -->
<img src="./photo.jpg" alt="Photo">

<!-- PNG - Transparent background -->
<img src="./logo.png" alt="Logo with transparent background">
```

```
<!-- SVG - Scalable images -->
<img src="./mario.svg" alt="Mario" height="100px">
```

Path Theory

Types of Paths

Absolute Path

The complete, unique path to a file from the root directory.

Examples:

- Windows: `C:\Users\John\Documents\photo.jpg`
- Mac/Linux: `/Users/John/Documents/photo.jpg`

Relative Path

The path created relative to your current working directory.

Relative Path Symbols:

Symbol	Meaning	Example
`./`	Current directory	`./image.jpg`
`../`	Parent directory (one level up)	`../class2.html`
`../../`	Two levels up	`../../folder/file.html`

Path Examples:

```
Current location: /project/html/

File in same folder:     ./image.jpg          or just  image.jpg
File in parent folder:   ../class2.html
File in subfolder:       ./images/photo.jpg
File two levels up:      ../../style.css
```

Semantic HTML

What is Semantic HTML?

Semantic HTML uses meaningful tags that describe the purpose of the content they contain. Instead of using generic `<div>` tags, semantic tags tell both browsers and developers what type of content they hold.

Use Cases of Semantic HTML

Benefit	Description
Accessibility	Screen readers can navigate and interpret content better
SEO (Search Engine Optimization)	Search engines understand page structure and rank content appropriately
Maintenance	Developers can easily read and understand code structure

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Common Semantic Tags

Tag	Purpose	Example
`<header>`	Contains introductory content, logos, navigation	`<header>...</header>`
`<nav>`	Contains navigation links	`<nav>...</nav>`
`<main>`	Main content of the page	`<main>...</main>`
`<footer>`	Contains footer content (contact, copyright)	`<footer>...</footer>`

Note: Other semantic tags like `<section>`, `<article>`, and `<aside>` will be covered later.

Semantic HTML Example: Travel Website

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Travel Website</title>
</head>

<body>
    <!-- Header Section -->
    <header>
        <h1>Travel Website</h1>
        <nav>
```

```
            <a href="">Manali</a>
            <a href="">Kasol</a>
            <a href="">Goa</a>
        </nav>
    </header>

    <!-- Main Content -->
    <main>
        <h2>Destinations</h2>
        <h3>Manali</h3>
        <img src="images.jpeg" alt="Manali" height="200px">
        <h3>Kasol</h3>
        <img src="images.jpeg" alt="Kasol" height="200px">
    </main>

    <!-- Footer Section -->
    <footer>
        <h2>Contact Us</h2>
        <ul>
            <li>Instagram</li>
            <li>Facebook</li>
            <li>Phone Number</li>
        </ul>
    </footer>
</body>

</html>
```

Page Structure Visualization:

```
+-----------------------------------------+
| <header>                                 |
|   <h1> Travel Website </h1>              |
|   <nav> Manali | Kasol | Goa </nav>      |
+-----------------------------------------+
| <main>                                   |
|   <h2> Destinations </h2>                |
|   <h3> Manali </h3>                      |
|   <img>                                  |
|   <h3> Kasol </h3>                       |
|   <img>                                  |
+-----------------------------------------+
| <footer>                                 |
|   <h2> Contact Us </h2>                  |
|   Instagram | Facebook | Phone           |
+-----------------------------------------+
```